UFC Strategic Event Planning Report

Section 1: Context and Problem Statement

Business Domain and Problem

The Ultimate Fighting Championship (UFC) operates in the global sports entertainment domain, where event location, fighter matchups, and audience engagement directly influence revenue and brand equity. Recent revenue declines in the first half of the year highlight inefficiencies in event planning and market prioritization.

Why the Problem Matters

Declining pay-per-view (PPV) sales and attendance rates threaten UFC's market share and long-term profitability. Poor event placement and unoptimized fight cards can lead to underperforming events and reduced sponsor confidence. The challenge is to identify cities that balance profitability, fan engagement, and operational feasibility through data-driven planning.

Key Questions

1. Which global cities present the strongest opportunities for high-ROI UFC events?
2. How can fight card composition and market selection optimize both fan engagement and revenue?
3. What investment balance (media, payouts, operations) yields the best returns per venue?

Scope - This report focuses on building a quantitative framework to guide venue selection, fight card structuring, and investment allocation. It covers past UFC events (1994 - 2021) and projected financial performance across five candidate cities - Las Vegas, Abu Dhabi, New York, London, and Rio de Janeiro.

Section 2: Data Overview

Data Sources: Kaggle UFC Master Dataset (1994–2021): Fight outcomes, event metadata, and athlete performance statistics.

Dataset Characteristics

- **Records:** ~ 6,000 fights across 500+ events
- **Variables:** 45+, including win rate, finishes, significant strikes, event city, PPV sales, and payouts
- **Time Period:** 1994 - 2021
- **Key Variables:**
  - *Win Rate* - fighter success ratio
  - *Total Fights* measures of experience
  - *Finishes (KO/TKO)* - measures of entertainment value
  - *Significant Strikes* - indicator of fight intensity
  - *Venue Popularity* - derived from attendance and PPV metrics

Data Preprocessing

- Missing values were imputed via mean substitution and verified through correlation analysis.
- Outliers were removed using statistical thresholds to maintain data reliability.
- Min-max normalization ensured variable comparability.

Data Limitations & Bias

- Incomplete proprietary data for recent events post-2021.
- Venue cost estimates vary due to non-public contractual data.

Section 3: Methodology

1. **Quantitative Screening:**
    - Regression analysis linked these variables to PPV sales and attendance.
    - Top-performing cities: *Las Vegas, Abu Dhabi, New York, London, Rio de Janeiro.*

	Fight-City	Composite Score
0	Las Vegas	100.0
1	Abu Dhabi	82.4
2	New York	64.0
3	London	48.0
4	Rio de Janeiro	33.0

2. **Strategic Validation:**
    - Alignment with UFC's global expansion priorities and operational feasibility.
    - Balanced established and emerging markets to manage revenue and brand growth.

Regression-Based Metric Weighting: A multivariate linear regression model determined the **Popularity Index (PI)** of the fighters.

- **Dependent Variable:** PPV sales and engagement score
- **Predictors:** Win rate, total fights, KO/TKO finishes, significant strikes, venue popularity
- **Derived Weights:** Weights do not total 100% as they reflect regression-derived importance, not percentage allocation
    - Win Rate - 35%
    - Total Fights - 25%
    - Finishes - 25%
    - Significant Strikes - 10%

	Location & Venue	Fight Card	Weight Class	Reward tier	Risk Tier
0	Madison Square Garden, New York	(Main) Jon Jones vs. Sergei Pavlovich	Heavyweight	High	High risk
1	Madison Square Garden, New York	(Co-main) Colby Covington vs. Shavkat Rakhmonov	Welterweight	Medium	Medium risk
2	Etihad Arena, Abu Dhabi	(Main) Khamzat Chimaev vs. Sean Strickland	Middleweight	High	High risk
3	Etihad Arena, Abu Dhabi	(Co-main) Islam Makhachev vs. Beneil Dariush	Lightweight	High	Safe bet
4	T-Mobile Arena, Las Vegas	(Main) Justin Gaethje vs. Rafael Fiziev	Lightweight	High	Medium risk
5	T-Mobile Arena, Las Vegas	(Co-main) Dustin Poirier vs. Arman Tsarukyan	Lightweight	High	Safe bet
6	The O2 Arena, London	(Main) Leon Edwards vs. Belal Muhammad	Welterweight	Medium	Medium risk
7	The O2 Arena, London	(Co-main) Tom Aspinall vs. Curtis Blaydes	Heavyweight	Medium	Safe bet
8	Jeunesse Arena, Rio de Janeiro	(Main) Amanda Nunes vs. Irene Aldana	Women's Bantamweight	Medium	Medium risk
9	Jeunesse Arena, Rio de Janeiro	(Co-main) Deiveson Figueiredo vs. Petr Yan	Flyweight vs Bantamweight	Medium	Safe bet

Section 4: Investment and ROI Modelling

- **Constrained Regression Optimization:** Balanced investment categories to maximize ROI while minimizing volatility.
- **Scenario Testing:** Validated financial assumptions through sports benchmarks and stress testing.

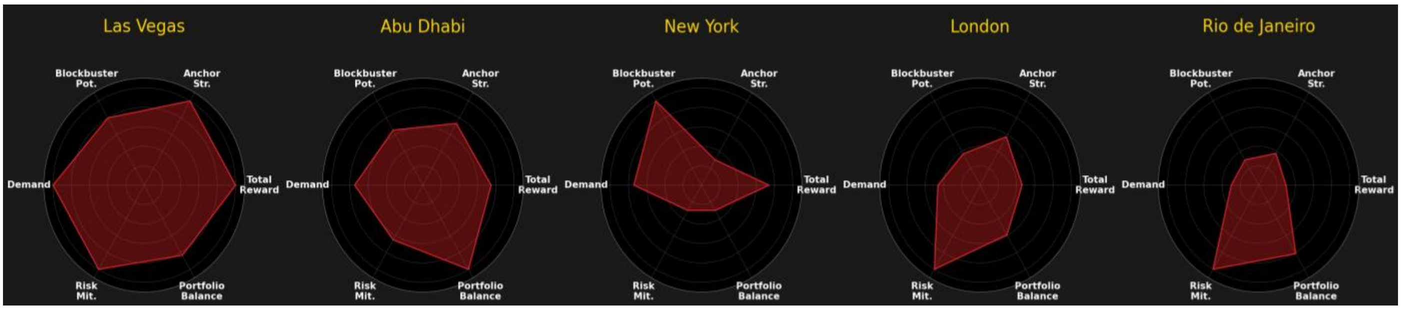
Location	Media Marketing	Fighter Payouts	Venue Operations	Total Investment	Expected ROI	Expected Return
Las Vegas	$8M	$20M	$15M	$43M	55%	$23.65M
Abu Dhabi	$4M	$12M	$8M	$24M	65%	$15.60M
New York	$7M	$18M	$12M	$37M	60%	$22.20M
London	$3M	$10M	$6M	$19M	68%	$12.92M
Rio de Janeiro	$2M	$8M	$4M	$14M	75%	$10.50M

Key ROI Insights

- **Rio de Janeiro:** Highest ROI (≈75%) due to low Opex and strong emerging market growth.
- **Las Vegas & New York:** High revenue potential but moderate ROI due to elevated expenses.
- **Abu Dhabi:** Strategic long-term growth market with favourable sponsorship dynamics.

Section 5: Killer Figure Design Process

- **Purpose:** Visualize and compare five proposed UFC event cities - Las Vegas, Abu Dhabi, New York, London, and Rio de Janeiro across six strategic dimensions.
- **Selected Visualization:** Multi-panel radar chart to show overall event performance and trade-offs between opportunity and risk.
- **Inclusions:** Six key decision dimensions - Total Reward, Anchor Strength, Blockbuster Potential, Demand, Risk Mitigation, and Portfolio Balance.
- **Exclusions:** Numeric values and secondary metrics excluded to focus on relative performance.
- **Narrative Role:** Summarizes city attractiveness at a glance and supports executive decision-making on event prioritization and resource allocation.



Early iterations used bar and heatmap visuals but proved less effective for comparing multiple variables. Based on feedback, axis labels, contrast, and readability were improved.

Section 6: Biggest Challenge

- **Type:** Data-related and technical.
- **Challenge:** Integrating and normalizing data from the dataset
- **Difficulty:** Inconsistent scales, missing values, and differing data quality across regions.
- **Approach:** Applied min-max normalization, imputation for missing data, and outlier removal.
- **Key Learning:** Robust preprocessing and version control are essential for reproducible analytics.